2023-24

S4 MATH CP

PAPER II

MC

Name: _____

Class: _____( )

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Four First Term Examination

MATHEMATICS Compulsory Part
PAPER 2

04/01/2024

11:05 am – 12:20 pm (1¼ hours)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first write your Name Class and Class Number and insert the information required in the spaces provided. No extra time will be given for writing your Name, Class and Class number after the 'Time is up' announcement.

2. When told to open this book, you should check that all questions are there. Look for the words ' **END OF PAPER**' after the last question.

3. All questions carry equal marks.

4. **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answer clearly; otherwise you will lose marks if the answers cannot be captured.

5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

6. No marks will be deducted for wrong answers.

There are 30 questions in Section A and 15 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.
Choose the best answer for each question.

Section A

1. $\left(\frac{1}{5}\right)^{998}(-5)^{999} =$

A. –5.
B. –0.2.
C. 0.2.
D. 5.

2. $\frac{(3x^3)^4}{9x^6} =$

A. $9x^6$.
B. $9x^{75}$.
C. $\frac{4}{3}x$.
D. $\frac{4}{3}x^6$.

3. If $2a + 3 = 2(b - 1)$, then $b =$

A. $a + \frac{1}{2}$.
B. $a + \frac{5}{2}$.
C. $a + 2$.
D. $a + 4$.

4. If $\frac{2x-3}{m} = \frac{3x+2}{n}$, then $x =$

A. $\frac{2n-3m}{3n+2m}$.
B. $\frac{3n+2m}{2n-3m}$.
C. $\frac{2n+3m}{3n+2m}$.
D. $\frac{3n+2m}{2n+3m}$.

5. Let a and b be constants. If $6x^2 + a(x + 2) – 9 \equiv 2ax(x + 2) + b(3x + 1)$, then b =

A. –9.

B. –3.

C. 3.

D. 6.

6. Let m and n be constants. If $2x^2 + m(x – 1) + 6 \equiv mx(x + 2) + n(x – 2)$, then n =

A. –3.

B. –2.

C. 2.

D. 4.

7. $36 – (3x – 2y)^2$ =

A. $(6 – 3x + 2y)(6 + 3x – 2y)$.

B. $(6 – 3x – 2y)(6 + 3x – 2y)$.

C. $(6 – 3x + 2y)(6 + 3x + 2y)$.

D. $(6 – 3x – 2y)(6 + 3x + 2y)$.

8. $x^2 + xy – 2y^2 + x – y$ =

A. $(x – y)(x – 2y + 1)$.

B. $(x – y)(x + 2y + 1)$.

C. $(x + y)(x – 2y – 1)$.

D. $(x + y)(x + 2y – 1)$.

9. If $x – 4y + 27 = 5x + y = 39$, then x =

A. –1.

B. 6.

C. 8.

D. 9.

10. The price of 5 shirts and 3 pairs of trousers is \$870. If the price of 3 shirts and the price of 4 pairs of trousers are the same, then the price of a shirt is

A. \$140.

B. \$130.

C. \$120.

D. \$90.

11. 0.0532646

A. 0.053 (correct to 3 significant figures).

B. 0.0532 (correct to 4 decimal places).

C. 0.05326 (correct to 5 significant figures).

D. 0.053265 (correct to 6 decimal places).

12. $\frac{1}{4\pi} =$

A. 0.080 (correct to 3 significant figures).

B. 0.07957 (correct to 4 significant figures).

C. 0.079577 (correct to 5 decimal places).

D. 0.079577 (correct to 6 decimal places).

13. Which of the following is a rational number?

A. 3.14π

B. $\sqrt{2}$

C. $\sqrt{9+16}$

D. $4\pi^2$

14. The fraction form of $0.0\dot{6}\dot{5}$ is

A. $\frac{65}{99}$.

B. $\frac{65}{111}$.

C. $\frac{13}{198}$.

D. $\frac{65}{999}$.

15. Which of the following equations has a repeated root?

A. $(x + 1)^2 = 8$

B. $(x - 3)^2 = 0$

C. $x^2 - 6x - 9 = 0$

D. $x^2 + 6x + 6 = 0$

16. If the straight line $2x + 5y + k = 0$ passes through $(-3, 2)$, then $k =$

A. -4.

B. -2.

C. 2.

D. 4.

17. Which of the following are equations of straight lines?

I. $x + 3 = 0$

II. $3x - 4y = 0$

III. $y - 4 = 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

18. Which of the following straight lines has a negative slope?

A. $y = \frac{2}{3}x - 5$

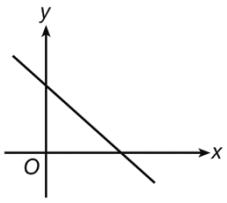
B. $2x - 5y = 10$

C. $12 - 2x - 3y = 0$

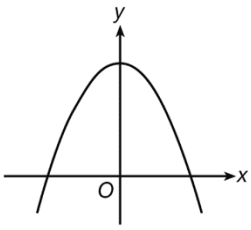
D. $y - 1 = -3(2 - x)$

19. Which of the following shows that y must **NOT** be a function of x?

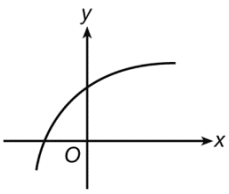
A.



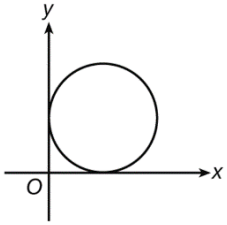
B.



C.



D.



20. Which of the following is **NOT** a function of x?

A. $y = x + 4$

B. $y = \sqrt{x} + 4$

C. $y = x^2 + 4$

D. $y^2 = x + 4$

21. If $f(x) = 5 - x - x^3$, then $f(-1) =$

A. 1.

B. 3.

C. 5.

D. 7.

22. Which of the following graphs of quadratic functions opens downwards?

A. $y = x^2 + 2x$

B. $y = x^2 - 5x - 6$

C. $y = -2x^2 + 5x$

D. $y = -5x + 2x^2$

23. It is given that the axis of symmetry of the graph of $y = -5x^2 + 20x$ is $x = 2$. The coordinates of the vertex of the graph are

A. (2, 20).

B. (2, −20).

C. (−2, 20).

D. (−2, −20).

24. Let $f(x) = 2(x-1)^2 + k$. If the graph of $y = f(x)$ passes through the point (2, 8), then the y-intercept of the graph of $y = f(x)$ is

A. 4.

B. 8.

C. 10.

D. 12.

25. Let a and b be constants. When $x^4 + ax^2 + b$ is divided by $x^2 - x + 3$, which of the following **CANNOT** be the remainder?

A. 5

B. $x - 3$

C. $x + 5$

D. $x^2 + 4x - 1$

26. When $x^3 + 2x^2 - 3x + 5$ is divided by $2 + x$, the remainder is

A. 11.

B. 5.

C. 3.

D. −5.

27. Let $P(x)$ be a polynomial. If $P\left(\frac{2}{3}\right) = 0$, which of the following must be a factor of $P(x)$?

A. $2x - 3$

B. $2x + 3$

C. $3x - 2$

D. $3x + 2$

28. Which of the following angles does **NOT** lie in quadrant II?

A. 130°

B. −190°

C. 420°

D. 530°

29. cos 185° =

A. −cos 5°.

B. −cos 15°.

C. cos 5°.

D. cos 15°.

30. If $\tan\theta = \sqrt{2} - \sqrt{3}$, then θ lies in

A. quadrant I or II only.

B. quadrant I or IV only.

C. quadrant II or III only.

D. quadrant II or IV only.

Section B

31. Let a be a constant. Solve the equation $x(x - 1) = a(a - 1)$.

A. $x = a$

B. $x = a$ or $x = 1 - a$

C. $x = a$ or $x = a + 1$

D. $x = a$ or $x = a - 1$

32. Let k be a constant. If the quadratic equation $x^2 - kx + 3k = 5$ has equal roots, then

A. $k = 10$.

B. $k = 12$.

C. $k = 0$ or $k = 12$.

D. $k = 2$ or $k = 10$.

33. If β is a root of the equation $x^2 - 3x + 1 = 0$, then $2\beta^2 - 6\beta - 1 =$

A. -3.

B. -1.

C. 0.

D. 1.

34. If the straight lines $hx + ky - 16 = 0$ and $2x + 5y + 8 = 0$ are perpendicular to each other and intersect at a point on the y-axis, then $h =$

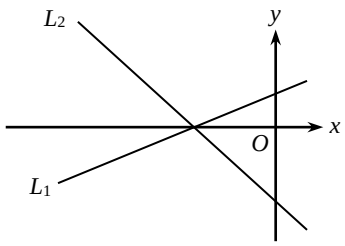
A. -25.

B. -10.

C. 10.

D. 25.

35. In the figure, the two straight lines L_1: $x + ay + b = 0$ and L_2: $mx + y + n = 0$ intersect at a point on the negative x-axis.



Which of the following are true?

I. $n = bm$ II. $a > 0$ III. $n > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

36. The coordinates of the points A and B are (3, -2) and (11, 6) respectively. Let L be the perpendicular bisector of AB. Find the equation of L.

A. $x + y - 8 = 0$

B. $x - y = 0$

C. $x + y - 9 = 0$

D. $x - y - 5 = 0$

37. Let $f(x) = 2x^2 - x - 3$. If β is a constant, then $f(1 - \beta) - f(1 + \beta) =$

A. -10β.

B. -6β.

C. $4\beta^2 - 4$.

D. $4\beta^2 - 8$.

38. If $f(x) = 3x^2 - x + 1$, then $f(2m - 1) =$

A. $6m^2 - 2m + 3$.

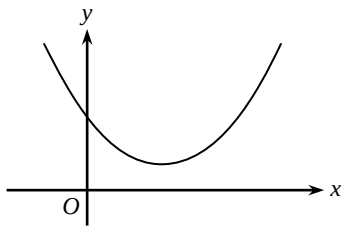
B. $6m^2 - 2m + 5$.

C. $12m^2 - 14m + 3$.

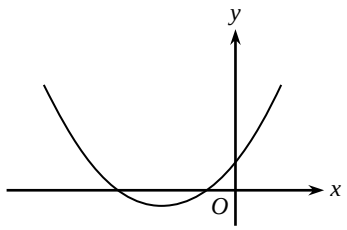
D. $12m^2 - 14m + 5$.

39. If $hk < 0$, which of the following may represent the graph of $y = (h - x)^2 + k$?

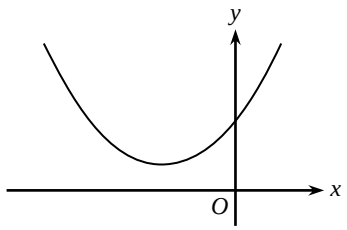
A.



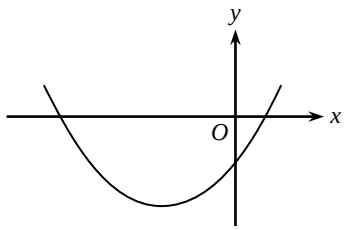
B.



C.



D.



40. Let $f(x) = 2x^2 + kx - 4$, where k is a constant. If the graph of $y = f(x)$ passes through the point $(1, -9)$, then the x-intercepts of the graph of $y = f(x)$ are

A. -4 and $\frac{1}{2}$.

B. -2 and $\frac{1}{2}$.

C. $-\frac{1}{2}$ and 4.

D. $-\frac{1}{2}$ and 2.

41. Let $f(x)$ be a polynomial. When $f(x)$ is divided by $x + 2$, the remainder is -8. If $f(x)$ is divisible by $x - 2$, find the remainder when $f(x)$ is divided by $x^2 - 4$.

A. $2x - 4$

B. $2x + 4$

C. $-2x - 4$

D. $-2x + 4$

42. Let $P(x)$ be a polynomial. When $P(x)$ is divided by $x + 2$, the remainder is R. In which of the following cases, the remainder must be R?

A. $P\left(\frac{x}{2}\right) \div (2x + 1)$

B. $P(x) \div (2x + 1)$

C. $P(2x) \div (2x + 1)$

D. $P(4x) \div (2x + 1)$

43. Let n be a positive odd number. Which of the following must be true?

I. $x^n - 1$ is divisible by $x - 1$.

II. $x^n + 1$ is divisible by $x + 1$.

III. $x^n + x^{n-1}$ is divisible by x.

A. I only

B. I and II only

C. II and III only

D. I, II and III

44. For $0° \le x \le 360°$, how many roots does the equation $2\cos^2 x = 3\cos x$ have?

A. 2

B. 3

C. 4

D. 5

45. The minimum value of $\dfrac{3}{4+2\sin x}$ is

A. $\dfrac{1}{2}$.

B. $\dfrac{3}{4}$.

C. 3.

D. $\dfrac{3}{2}$.

END OF PAPER